

Power

A. List ALL major components (excluding passives & regulators)

Component Name	Part Number	Supply (V)
Microcontroller	PIC18F57Q43	3.3
OLED Display	SH1106	3.3
Temp/Humidity Sensor (heated)	HDC2080	3.3
Programmer (target not powered)	MPLAB Snap	3.3
Status LED	LED	3.3
User LEDs (2)	LED x2	3.3
Push Button (pull-up current)	-	3.3

B. 3.3V Regulator Selection

Component Name	Part Number	Input Range
Buck Switching Regulator	AP63203WU-7	3.8V–32V

C. External Power Source

Component Name	Part Number	Output (V)
9V AC-DC Wall Adapter	WSU090-2500	9

Budget - PIC18F57Q43 System

#	Absolute (mA)	Total (mA)
1	30	30
1	100	100
1	90	90
1	0	0
1	10	10
2	3	6
1	0.33	0.33

Subtotal (mA) 236.33

Safety Margin (20%) 47.266

Total Current Required on +3 283.596

Max Output (mA)
2000
Remaining Current Available 1716.404

Max Current (mA)
2500
Estimated Input Current (mA) 122.3355294
Remaining Current on Adapter 2377.664471



Unit
mA
mA
mA
mA
mA
mA
mA

